

Foundations of Natural language processing

[Natural language processing]

1.0 Content Block

Grammar induction Generate a formal grammar that describes a language's syntax. **Sentence breaking** (also known as "sentence boundary disambiguation") Given a chunk of text, find the sentence boundaries. Sentence boundaries are often marked by periods or other punctuation marks, but these same characters can serve other purposes (e.g., marking abbreviations). **Parsing** Determine the parse tree (grammatical analysis) of a given sentence. The grammar for natural languages is ambiguous and typical sentences have multiple possible analyses: perhaps surprisingly, for a typical sentence there may be thousands of potential parses (most of which will seem completely nonsensical to a human). There are two primary types of parsing: dependency parsing and constituency parsing. Dependency parsing focuses on the relationships between words in a sentence (marking things like primary objects and predicates), whereas constituency parsing focuses on building out the parse tree using a probabilistic context-free grammar (PCFG) (see also stochastic grammar).

2.0 Content Block

Apply the theory of conceptual metaphor, explained by Lakoff as "the understanding of one idea, in terms of another" which provides an idea of the intent of the author. For example, consider the English word big. When used in a comparison ("That is a big tree"), the author's intent is to imply that the tree is physically large relative to other trees or the author's experience. When used metaphorically ("Tomorrow is a big day"), the author's intent is to imply importance. The intent behind other usages, like in "She is a big person", will remain somewhat ambiguous to a person and a cognitive NLP algorithm alike without additional information. **Assign relative measures of meaning to a word, phrase, sentence or piece of text based on the information presented before and after the piece of text being analyzed**, e.g., by means of a probabilistic context-free grammar (PCFG). The mathematical equation for such algorithms is presented in US Patent 9269353:

3.0 Content Block

Statistical approach In the late 1980s and mid-1990s, the statistical approach ended a period of AI winter, which was caused by the inefficiencies of the rule-based approaches. The earliest decision trees, producing systems of hard if-then rules, were still very similar to the old rule-based approaches. Only the introduction of hidden Markov models, applied to part-of-speech tagging, announced the end of the old rule-based approach.

4.0 Content Block

Lemmatization The task of removing inflectional endings only and to return the base dictionary form of a word which is also known as a lemma. Lemmatization is another technique for reducing words to their normalized form. But in this case, the transformation actually uses a dictionary to map words to their actual form. **Morphological segmentation** Separate words into individual morphemes and identify the class of the morphemes. The difficulty of this task depends greatly on the complexity of the morphology (i.e., the structure of words) of the language being considered. English has fairly simple morphology, especially inflectional morphology, and thus it is often possible to ignore this task entirely and simply model all possible forms of a word (e.g., "open, opens, opened, opening") as separate words. In languages such as Turkish or Meitei, a highly agglutinated Indian language, however, such an approach is not possible, as each dictionary entry has thousands of possible word forms. **Part-of-speech tagging** Given a sentence, determine the part of speech (POS) for each word. Many words, especially common ones, can serve as

multiple parts of speech. For example, "book" can be a noun ("the book on the table") or verb ("to book a flight"); "set" can be a noun, verb or adjective; and "out" can be any of at least five different parts of speech. Stemming The process of reducing inflected (or sometimes derived) words to a base form (e.g., "close" will be the root for "closed", "closing", "close", "closer" etc.). Stemming yields similar results as lemmatization, but does so on grounds of rules, not a dictionary.

5.0 Content Block

Natural language processing (NLP) is the processing of natural language information by a computer. NLP is a subfield of computer science and is closely associated with artificial intelligence. NLP is also related to information retrieval, knowledge representation, computational linguistics, and linguistics more broadly. Major processing tasks in an NLP system include: speech recognition, text classification, natural language understanding, and natural language generation.

6.0 Content Block

The premise of symbolic NLP is often illustrated using John Searle's Chinese room thought experiment: Given a collection of rules (e.g., a Chinese phrasebook, with questions and matching answers), the computer emulates natural language understanding (or other NLP tasks) by applying those rules to the data it confronts.

7.0 Content Block

Discourse (semantics beyond individual sentences) Coreference resolution Given a sentence or larger chunk of text, determine which words ("mentions") refer to the same objects ("entities"). Anaphora resolution is a specific example of this task, and is specifically concerned with matching up pronouns with the nouns or names to which they refer. The more general task of coreference resolution also includes identifying so-called "bridging relationships" involving referring expressions. For example, in a sentence such as "He entered John's house through the front door", "the front door" is a referring expression and the bridging relationship to be identified is the fact that the door being referred to is the front door of John's house (rather than of some other structure that might also be referred to). Discourse analysis This rubric includes several related tasks. One task is discourse parsing, i.e., identifying the discourse structure of a connected text, i.e. the nature of the discourse relationships between sentences (e.g. elaboration, explanation, contrast). Another possible task is recognizing and classifying the speech acts in a chunk of text (e.g. yes–no question, content question, statement, assertion, etc.). Implicit semantic role labelling Given a single sentence, identify and disambiguate semantic predicates (e.g., verbal frames) and their explicit semantic roles in the current sentence (see Semantic role labelling above). Then, identify semantic roles that are not explicitly realized in the current sentence, classify them into arguments that are explicitly realized elsewhere in the text and those that are not specified, and resolve the former against the local text. A closely related task is zero anaphora resolution, i.e., the extension of coreference resolution to pro-drop languages. Recognizing textual entailment Given two text fragments, determine if one being true entails the other, entails the other's negation, or allows the other to be either true or false. Topic segmentation and recognition Given a chunk of text, separate it into segments each of which is devoted to a topic, and identify the topic of the segment. Argument mining The goal of argument mining is the automatic extraction and identification of argumentative structures from natural language text with the aid of computer programs. Such argumentative structures include the premise, conclusions, the argument scheme and the relationship between the main and subsidiary argument, or the main and counter-argument within discourse.